

Building User Features and Segmentation

Behavioral Intelligence Report from Transactional Data

Dataset: Indonesian Digital Payment Platform Transactions
Period: March 1 – May 31, 2024 | 278,932 transactions | 24,321 users
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1. Objective

Raw transaction logs are one of the most underutilized assets in financial services. Every purchase encodes behavioral signals about lifestyle, financial discipline, and life stage. This report documents a behavioral intelligence pipeline that converts raw transaction data into user-level features, infers latent demographic and personality attributes, and segments users into actionable groups.

The pipeline addresses two goals defined in the task:

- Feature Creation: derive user-level features from transaction-level data across 7 behavioral dimensions
- User Segmentation: identify distinct user groups using both algorithmic (KMeans, GMM) and rule-based non-traditional methods

All demographic inferences in this report are probabilistic behavioral estimates derived from population-level patterns. They are not factual labels and should not be used for individual-level decisions.

2. Dataset Overview

Attribute	Value
Total transactions	278,932
Unique users	24,321
Date range	March 1 – May 31, 2024 (92 days)
Null values	0 (clean dataset)
Transaction amount range	Rp 1,000 – Rp 587,100 (mean Rp 50,098)
MCC codes	21 unique codes, mapped to 18 categories (VISA codes)
Geographic coverage	Jakarta, Surabaya, Yogyakarta
Payment methods	Balance (majority), Credit Card (~20%)
Device types	Android (various versions), iPhone, iPad

Key Data Limitation

The transaction_date column contains date only, contrary to the dataset description which states "date and time". **No time component exists** in any of the 278,932 rows. This eliminates all hour-based behavioral signals such as late-night ratio, work-hour ratio, and lunch-hour patterns, which are among the strongest lifestyle indicators in consumer transaction research.

Compensation strategy: we replaced hour-based features with richer date-level signals: payday window activity (25th–3rd of month), weekday diversity (spread of activity across days of week), and monthly spending consistency across the 3-month window.

3. Feature Engineering

All 278,932 transaction rows are aggregated to user level, producing a table of 24,321 users × 53 features. Features are organized into 7 groups. Every feature has an explicit behavioral justification.

Feature Engineering Pipeline

Transaction Data (278,932 rows) → User-Level Aggregation (24,321 users) → [5A] [5B] [5C] [5D] [5E] [5F] [5G] → 53 features → Inference + Segmentation

5A. Spending Features

Captures scale and volatility of user spending behavior.

Feature	Formula	Behavioral Signal
total_spend	Sum of all amounts	Overall economic footprint on platform
avg_spend	Mean transaction amount	Typical purchasing power per transaction
median_spend	Median transaction amount	Resistant to outlier distortion
spend_std	Std deviation of amounts	Spending volatility
cv_spend	std / mean	Normalized volatility: high = erratic, low = disciplined
transaction_count	Count of transactions	Platform engagement depth

5B. Temporal Features

Captures rhythm of spending across time. No hour-level data available.

Feature	Definition	Behavioral Signal
weekend_ratio	Share of weekend transactions	High = social/family lifestyle; Low = work-oriented
payday_spike_ratio	Share in payday window (25th–3rd)	High = salary-dependent; Low = stable cash flow
active_days	Distinct calendar days with transactions	Engagement regularity
tx_per_active_day	Transactions per active day	Intensity of usage
weekday_diversity	Std of weekday distribution	High = irregular schedule; Low = routine
monthly_spend_std	Std of monthly totals	High = volatile income or behavior

5C. Merchant Category Features

MCC codes are mapped to 18 categories. Per-user ratios reveal consumption priorities and lifestyle. Merchant and category diversity capture exploration tendency.

Category	Signal Interpretation
restaurant_ratio, fast_food_ratio	Food spending priority; extraversion proxy
grocery_ratio	Domestic/family-centered lifestyle
pharmacy_ratio	Health engagement; mature life stage signal
gaming_ratio	Younger, digital-native user
education_ratio, books_ratio	Student or professional development
fuel_ratio	Vehicle ownership proxy
beauty_ratio, fashion_ratio	Appearance-oriented; gender proxy component
travel_ratio, electronics_ratio	Higher income proxy
furniture_ratio, hardware_store_ratio	Homeownership and settled lifestyle
merchant_diversity	Unique merchant count
category_diversity	Unique MCC labels (lifestyle variety)

5D. Geo Features

Home location is estimated as the median lat/lon of weekend transactions. Work location is estimated from weekday transactions. Without time-of-day data, work coordinates cannot be isolated to work hours; weekday median is the closest available proxy.

Feature	Method	Signal
home_lat, home_lon	Median coordinates on weekends	Estimated residence location
home_city	Bounding box assignment	Jakarta, Surabaya, Yogyakarta, or Other
work_lat, work_lon	Median coordinates on weekdays	Estimated work area (proxy only)
mobility_radius	Combined std of lat/lon	Geographic spread: high = mobile/traveler

5E. Financial Behavior Features

Feature	Signal
discount_ratio	Price sensitivity: high = deal-seeker or budget-constrained
loyalty_ratio	Brand commitment and spending organization
refund_ratio	Indecisiveness or impulse purchasing behavior
failed_tx_ratio	Payment stress or insufficient balance signal
notes_ratio	Expense tracking behavior: conscientiousness proxy
avg_promo_amount	Magnitude of promotional savings per transaction
avg_merchant_rating	Quality preference or rating tendency

5F. Digital Sophistication Features

Feature	Signal
ios_ratio	iOS users empirically associated with higher purchasing power
credit_card_ratio	Requires bank approval; signals financial maturity (~20% of users)
avg_android_ver	Newer version = more tech-engaged user
device_score	Composite: $\text{ios_ratio} \times 10 + \text{android_version} \times 0.5$

5G. Big Five Personality Proxy Scores

Behavioral proxies inspired by consumer psychology literature. These are data-driven behavioral estimates, not clinical measurements. Each score is a weighted sum of normalized [0,1] features. Late-night ratio was excluded due to missing time data.

Trait	Key Behavioral Inputs	High Score Interpretation
Openness	merchant_diversity, category_diversity, travel_ratio, mobility_radius	Curious, variety-seeking, exploratory
Conscientiousness	loyalty_ratio, notes_ratio, low failed_tx, low cv_spend	Disciplined, organized, routine-driven
Extraversion	restaurant_ratio, fast_food_ratio, beauty_ratio, travel_ratio	Social, outward-facing, experience-seeking
Agreeableness	grocery_ratio, loyalty_ratio, avg_merchant_rating	Family/community-oriented, cooperative

Neuroticism	cv_spend, failed_tx_ratio, refund_ratio, payday_spike_ratio	Erratic, impulsive, financially stressed
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4. Human Attribute Inference

All inferences convert behavioral signals into probabilistic demographic estimates. No ground truth exists for validation. Thresholds are set at natural histogram break points in the score distributions, not at arbitrary quantiles. This section documents the methodology for each required attribute.

4.1 Income Level (1st Priority)

Income score is an average of 8 normalized spending power signals: avg_spend, total_spend, credit_card_ratio, device_score, mobility_radius, travel_ratio, electronics_ratio, and furniture_ratio. The distribution is roughly bell-shaped (min=0.026, median=0.211, max=0.448) with most users clustered in a narrow middle band. Four tiers are not supported by the signal spread; three tiers are used.

Level	Threshold	Users	Share
Low	score < 0.15	3,825	15.7%
Middle	0.15 – 0.27	14,873	61.2%
High	score > 0.27	5,623	23.1%

4.2 Age Group (1st Priority)

Age is inferred using balanced young and mature signal scores, each averaged across their components before subtracting, preventing structural imbalance. Young signals: gaming, fast food, books, weekend ratio, payday spike, low avg_spend. Mature signals: pharmacy, grocery, furniture, hardware, high avg_spend.

Caveat: MCC correlations with age are near-zero (maximum $r=0.009$ across all features). Age inference is the weakest attribute in this dataset. The right-skew toward younger groups reflects this platform genuinely attracting younger adults.

Age Group	Score Range	Users	Share
50+	score < -0.034	426	1.8%
35–50	-0.034 to 0.105	8,979	36.9%
25–34	0.105 to 0.244	13,720	56.4%
18–24	score > 0.244	1,196	4.9%

4.3 Working Status (1st Priority)

Inferred from weekday/weekend ratio and commute distance (gap between home and weekday median coordinates). Without time-of-day data (hour in dataset), work hours cannot be isolated. Classification applies in priority order: first match wins.

Status	Users	Share	Key Signals
Office Worker	13,852	57.0%	Weekday-dominant AND detectable commute gap
Home Based	4,988	20.5%	No weekday dominance, no student signals
Freelancer	2,155	8.9%	High mobility AND high weekday diversity
Professional	2,048	8.4%	Weekday-dominant, no detectable commute
Student	1,278	5.3%	Low spend AND education/books spending

4.4 Gender Proxy (1st Priority)

Inferred from culturally-gendered MCC categories for Indonesian consumers. Female signals: beauty_ratio, fashion_ratio. Male signals: gaming_ratio, sports_ratio, fuel_ratio. This proxy carries significant uncertainty and should be treated directionally only.

Gender Proxy	Users	Share
Likely Male	11,806	48.5%
Likely Female	7,252	29.8%
Indeterminate	5,263	21.6%

4.5 Educational Background (1st Priority)

edu_score combines education_ratio (weight 0.4), books_ratio (0.3), income_score (0.2), and device_score (0.1). Only 20% of users have any education merchant spend and 28% have any books spend. The score is right-skewed; three tiers are used. Distinguishing graduate from postgraduate is not supported by the signal.

Level	Threshold	Users	Share
High School or Below	score < 0.094	5,579	22.9%
Undergraduate	0.094 – 0.27	15,522	63.8%
Postgraduate	score > 0.27	3,220	13.2%

4.6 Home Location & Work Location (1st Priority)

Home location: median coordinates of weekend transactions per user. Work location: median coordinates of weekday transactions per user (proxy only, no time data available to isolate work hours). City assignment uses coordinate bounding boxes.

Home City	Users	Share
Jakarta	14,529	59.8%
Surabaya	4,695	19.3%
Yogyakarta	1,857	7.6%
Other / Unknown	2,276	9.4%

4.7 Marital & Parental Status (2nd Priority)

Family score combines grocery (0.3), education (0.3), pharmacy (0.2), weekend ratio (0.1), furniture (0.1). Marital proxy splits at the distribution median (0.122), a principled threshold for a binary label. Parental proxy uses two natural signals: any education merchant spend (20% of users have this) or top-quarter family score.

Attribute	Label	Users	Share
Marital	Likely Married	12,206	50.2%
Marital	Likely Single	12,115	49.8%
Parental	Likely Parent	8,159	33.5%
Parental	Likely No Children	16,162	66.5%

4.8 Vehicle Ownership (2nd Priority)

Vehicle score = fuel_ratio (0.7) + commute_dist (0.3). The distribution has a sharp cliff at 0.108 because fuel_ratio is zero for most users, this natural break point is the threshold. 24% vehicle ownership is plausible for urban Indonesian mobile payment users.

Label	Users	Share
Likely Owns Vehicle	5,786	23.8%
Likely No Vehicle	18,535	76.2%

4.9 Industry of Employment (2nd Priority)

Inferred from the dominant MCC spending category per user. This is the weakest inference: people who eat at restaurants most often are not necessarily in food service. The food_beverage dominance (61%) reflects restaurants being the most common MCC in the dataset, not actual employment distribution. Use directionally only.

Industry Proxy	Users	Share
Food & Beverage	14,778	60.8%
Retail (FMCG)	4,502	18.5%
Healthcare	1,453	6.0%
Technology	810	3.3%
Other (8 categories)	776	3.2%

4.10 Health Status (2nd Priority)

Health score = pharmacy_ratio (0.6) + sports_ratio (0.3) + financial stability (0.1). Labeled by engagement level, not health outcome. High pharmacy spend may indicate chronic conditions or proactive health management, both map to "health_conscious" in behavioral terms.

Status	Threshold	Users	Share
Low Engagement	score < 0.115	6,517	26.8%
Moderate	0.115 – 0.279	15,097	62.1%
Health Conscious	score > 0.279	2,707	11.1%

4.11 Homeownership Status (2nd Priority)

Homeownership score = furniture_ratio (0.35) + hardware_store_ratio (0.35) + income_score (0.2) + low mobility (0.1). Threshold at 0.214 corresponds to the histogram inflection point where home-improvement spending becomes meaningful.

Label	Users	Share
Likely Homeowner	7,118	29.3%
Likely Renter	17,203	70.7%

5. User Segmentation

5.1 Methodology

Two complementary segmentation approaches are applied:

- KMeans clustering (algorithmic): partitions users into k groups by minimizing within-cluster variance across 31 standardized behavioral features. Best k selected by silhouette score.
- Rule-based behavioral archetypes (non-traditional): deterministic classification using explicit behavioral rules. No distance metric, no random seed. Addresses the task's extra-points criterion for segmentation beyond traditional methods.

5.2 Feature Selection for Clustering

31 raw behavioral features are used as clustering inputs. Inferred labels (income_level, age_group, etc.) are excluded to prevent circular reasoning. Psychological scores are excluded because they are derived from the same raw features, which would create redundancy and compress cluster variance.

5.3 Standardization and Dimensionality Reduction

StandardScaler transforms each feature to zero mean and unit variance before clustering, preventing features with large ranges (e.g. total_spend in IDR) from dominating the distance calculation. PCA reduces to 2 components for visualization only; clustering is performed in the full 31-dimensional space.

PCA explained variance: 19.46% across 2 components. The low explained variance confirms the data is genuinely high-dimensional without dominant linear structure, which partly explains the low silhouette scores.

5.4 Cluster Count Selection

Silhouette score was computed for k=3 to k=8 using a 5,000-user sample:

k	Silhouette Score
3	0.0321
4	0.0386 (best)
5	0.0350
6	0.0320
7	0.0330
8	0.0329

All silhouette scores are low (< 0.04). This is an honest finding: this platform's users are behaviorally homogeneous across most dimensions. The dataset lacks the strong natural cluster structure that would produce high silhouette scores. $k=4$ is selected as it achieves the highest score while remaining interpretable.

5.5 GMM Comparison

Gaussian Mixture Models were evaluated as an alternative. GMM assigns probabilistic cluster membership rather than hard assignments, which is theoretically more appropriate when clusters overlap. BIC (Bayesian Information Criterion) selected $k=7$ as the optimal GMM solution.

However, the GMM result produced one dominant cluster of 11,560 users with an average assignment confidence of 0.998. High confidence in a low-silhouette dataset indicates the model found one large behavioral blob, not nuanced structure. KMeans with $k=4$ is preferred for interpretability.

5.6 Cluster Profiles

Metric	Active Explorers	Established Regulars	Gaming Casuals	Food-Focused Occ.
Users	5,524 (22.7%)	10,302 (42.4%)	1,741 (7.2%)	6,754 (27.8%)
Avg Spend (IDR)	54,246	50,996	48,849	45,583
Avg Transactions	15.9	11.6	10.9	7.8
Active Days	14.5	10.8	10.3	7.5
Merchant Diversity	15.8	11.6	10.9	7.8
Loyalty Ratio	0.228	0.246	0.254	0.241
Discount Ratio	0.173	0.171	0.174	0.167
Credit Card Ratio	0.199	0.197	0.203	0.208
Gaming Ratio	0.008	0.000	0.102	0.000
Restaurant Ratio	0.231	0.239	0.223	0.257
Pharmacy Ratio	0.095	0.096	0.087	0.096
Openness Score	0.424	0.334	0.334	0.246
Conscientiousness	0.475	0.484	0.488	0.485
Neuroticism	0.225	0.222	0.218	0.212

6. Segment Descriptions & Business Recommendations

Segment 1: Active Explorers (5,524 users, 22.7%)

Active Explorers are the platform's highest-value, most engaged users across every measurable dimension. They spend the most per transaction (avg IDR 54,246), transact most frequently (15.9 transactions per month), are active the most days per month (14.5), and visit the widest range of merchants (15.8 unique). Their openness score (0.424) is the highest of all segments, confirming they seek variety and explore new merchants readily. Electronics is their distinguishing MCC category, pointing to tech-savvy spending behavior.

- **Inferred demographic profile:** skewed toward high income (33%), predominantly 25–34 age group (56%), likely male (57%), office workers and professionals (67% combined).
- **Personality profile:** high openness, moderate conscientiousness. Curious and active, but not the most disciplined, they transact frequently and widely but show lower loyalty ratio than other segments (0.228).
- **Distinguishing MCC:** electronics_ratio (0.051) and misc_retail_ratio (0.050) are highest in this cluster, alongside strong performance across all food categories.
- **Business opportunity:** Premium membership tiers, curated discovery features, and cross-category recommendations are the highest-leverage actions. This segment has the greatest lifetime value potential. Even small retention improvements generate outsized revenue impact. Reward breadth of spending, not just volume. Consider early access programs and personalized merchant discovery feeds.

Segment 2: Established Regulars (10,302 users, 42.4%)

Established Regulars are the platform's largest and most stable segment. Moderate spend (IDR 50,996), moderate frequency (11.6 transactions), moderate diversity (11.6 merchants). What distinguishes them is their MCC profile: highest pharmacy (0.096), fuel (0.049), and fashion ratios (0.043) of all clusters, a spending pattern consistent with working adults managing health, commuting, and personal appearance.

- **Inferred demographic profile:** mix of 25–34 (56%) and 35–50 (39%), predominantly middle income (62%), office workers majority (57%), with the lowest male skew of all segments (46% male, 34% female).
- **Personality profile:** highest conscientiousness score (0.484). Routine-driven, organized, health-aware. Their spending is predictable and purposeful, not impulsive.
- **Business opportunity:** Loyalty program deepening, insurance products, health-related financial products, and cashback on pharmacy and fuel. This segment is the platform's retention backbone — they will not churn easily but will

not upgrade without a clear value proposition. Focus on stickiness and cross-sell into financial products that match their established-adult lifestyle.

Segment 3: Gaming Casuals (1,741 users, 7.2%)

Gaming Casuals are the smallest and most distinctive segment. Their only standout characteristic is gaming_ratio (0.102, essentially zero in all other clusters). Every other behavioral dimension is average: spend (IDR 48,849), frequency (10.9 transactions), active days (10.3). They are not outliers in spending scale, they are outliers in category concentration.

- **Inferred demographic profile:** strongly male (94%, the highest male skew of all segments), youngest-skewing within segments (15% are 18–24 vs 1–3% in other segments), lower income proportion (16% low income). The indeterminate gender proportion is 0%, all users in this segment have a clear male or female behavioral signal.
 - **Personality profile:** moderate neuroticism, lower agreeableness. Digital-native orientation but narrow, their interest is concentrated, not broad.
 - **Business opportunity:** In-app gaming integrations, digital entertainment vouchers, and gaming merchant partnerships. This segment has a strong identity and responds well to targeted campaigns that speak directly to their gaming lifestyle. Avoid generic campaigns. The 18–24 concentration makes them a long-term acquisition priority: their spending power will grow as they age into higher income brackets.
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Segment 4: Food-Focused Occasionals (6,754 users — 27.8%)

Food-Focused Occasionals are the least active segment by every transactional measure: lowest spend (IDR 45,583), lowest frequency (7.8 transactions), fewest active days (7.5). When they do transact, it is almost exclusively food: highest restaurant (0.257), fast food (0.141), and grocery ratios (0.151) of all clusters. Their merchant diversity is the lowest (7.8), reflecting a narrow, food-centric footprint. Openness score (0.246) is the lowest of all segments.

- **Inferred demographic profile:** youngest-leaning alongside Gaming Casuals — 69% in 25–34, 8% in 18–24. Low-to-middle income (84% combined). Highest food_beverage industry proxy (dominant restaurant spending). Gender distribution is most ambiguous (39% indeterminate), suggesting this segment uses the platform primarily as a food payment tool without strong lifestyle differentiation.
 - **Personality profile:** low openness, moderate conscientiousness. Convenience and routine-oriented. They use the platform instrumentally, not as a financial lifestyle hub.
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- **Business opportunity:** Food delivery partnerships, restaurant cashback, and grocery rewards are the immediate hooks. The activation goal is frequency, nudge users beyond their food comfort zone with low-friction adjacent offers (e.g. pharmacy cashback alongside grocery purchases). Converting 20% of this segment to broader platform engagement would meaningfully shift their lifetime value.

7. Non-Traditional Segmentation: Behavioral Archetypes

The task awards extra points for segmentation beyond traditional methods. This section implements rule-based behavioral archetypes as a complementary layer. Unlike KMeans, this approach is fully interpretable: each archetype is defined by explicit behavioral rules, produces deterministic results (no distance metric, no random seed), and can be explained in plain language to any stakeholder.

Archetype	Users	Share	Defining Rule
General User	11,835	48.7%	Does not meet any specific archetype criteria
Occasional User	4,259	17.5%	active_days below 25th percentile (< 7 days)
Stable Family Earner	4,180	17.2%	High family score AND weekend_ratio > 0.3
Loyal Power User	2,808	11.5%	loyalty_ratio > 0.5 AND high transaction count
Deal Hunter	1,239	5.1%	discount_ratio > 0.3 AND payday_spike_ratio > 0.3

Archetype Profiles

Loyal Power Users (2,808 users)

Every second transaction is on a loyalty program AND they are above-median frequency users. These are the most retention-stable users on the platform. Action: VIP tier, early product access, referral program. Protect at all costs, losing a Loyal Power User is 1.5× more damaging than losing an average user.

Deal Hunters (1,239 users)

More than 30% of their transactions have discounts AND they cluster their spending in the payday window. Highly price-sensitive and salary-dependent. Action: salary-day flash sales, cashback campaigns timed to the 25th–1st window. Do not compete on value alone, these users respond to time-limited price triggers.

Stable Family Earners (4,180 users)

Strong household signals (grocery, education, pharmacy, furniture) combined with weekend-heavy activity. They use the platform for family provisioning. Action: family bundles, insurance products, education savings products. Cross-sell should emphasize protection and planning, not discovery.

Occasional Users (4,259 users)

Fewer than 7 active days across a 3-month window. The platform is not a habit for these users. Action: re-engagement push notifications, win-back offers, frictionless onboarding flows. The re-engagement window is highest in the first 30 days of inactivity.

8. Methodology Notes & Assumptions

Threshold Selection Philosophy

All inference thresholds are set at natural break points in the actual score distributions, derived from histogram inspection. Quantile-based binning was explicitly avoided because it forces equal group sizes regardless of the underlying data shape, producing labels that are mathematically tidy but behaviorally dishonest. For example, quartile binning on `income_score` would assign 25% of users to "affluent" even though the signal does not support that distinction.

Known Limitations

- **Low silhouette scores (0.032–0.039):** cluster boundaries are soft. Users near centroids are well-assigned; users at boundaries are ambiguous. Treat segment membership as probabilistic.
- **3-month window:** seasonal patterns, annual income cycles, and life-event spending (weddings, school enrollment) are invisible. A 12-month window would produce substantially richer temporal features.
- **No time-of-day data:** work-hour and late-night behavioral signals are entirely absent due to the dataset limitation despite the description claiming otherwise.
- **MCC correlation with demographics is near-zero** (max $r=0.009$): age, income, and education inferences are directional estimates only.
- **City differences are minimal:** all three cities show nearly identical avg spend (~Rp 50,000) and credit card rate (~20%). City-level segmentation models are unlikely to add value with this data.
- Industry proxy is **dominated by food_beverage** (61%) because restaurants are the most common MCC, not because most users work in food service.

Future Improvements

- Validate inferred demographics against opt-in survey data to calibrate proxy accuracy
- Extend to a 12-month transaction window for seasonal and lifecycle signals (if possible)
- Apply NLP to merchant names, raw names contain rich unlabeled category signals beyond MCC codes
- Use UMAP instead of PCA for dimensionality reduction, better preserves local structure at low silhouette scores
- Build ensemble segmentation: combine KMeans clusters with rule-based archetypes into a unified user profile score
- Per-city recalibration if city-level behavioral differences emerge in a larger dataset

9. Ethical Considerations

Demographic inference from transactional data carries inherent risks. The following considerations apply to all outputs in this report.

- Probabilistic, not factual. All inferred attributes are behavioral estimates, not verified facts. They reflect population-level correlations, not individual certainty. A user labeled "likely_male" may be female.
- Cultural and socioeconomic bias. Behavioral proxies are calibrated for Indonesian consumer patterns. Applying them to other markets without recalibration will introduce systematic errors.
- Self-reinforcing loops. Using behavioral estimates to target users with certain products can reinforce the spending patterns used to infer the estimate, limiting user diversity over time.
- Fairness constraints. Inferred attributes must not be used for discriminatory decisions: denying credit, charging differential prices, or limiting access to services based on inferred demographics.
- Privacy. Geo-derived home and work coordinates are sensitive personal data. Store with appropriate access controls; never expose at individual level in reports or APIs.
- Gender proxy limitations. Gender inference from MCC categories is particularly weak and culturally contingent. The 21.6% indeterminate rate reflects genuine ambiguity. This proxy should not be used in any decision that could disadvantage users on the basis of gender.

These outputs are intended to inform product strategy and campaign design at population level, not to make individual-level determinations about any specific user.